HEALTH SECTOR CAPSTONE PROJECT

INTRODUCTION

For this capstone project, you are working as a team of Junior Data Scientists for a healthcare-focused analytics consulting engagement. You have been assigned to a health insurance analytics company known as UrbanNest Health Analytics. The company works with insurance providers to help them better understand how patient characteristics influence insurance costs.

Currently, the company is facing a challenge: insurance charges are not being estimated consistently. Pricing is often influenced by general assumptions rather than data-driven insights. This has led to cases where some clients are overcharged while others are undercharged, affecting both customer satisfaction and company profitability.

Your responsibility as a data science team is to build a simple Linear Regression model that can predict insurance charges based on patient attributes. The purpose of this project is to give you hands-on experience with the full data science workflow, from data understanding to model evaluation, using a real-world healthcare dataset.

DATASET

You are required to use the **Medical Cost Personal Dataset**, which is a standard beginner-friendly dataset for regression problems in healthcare analytics. It contains information about patients such as age, gender, BMI, number of children, smoking status, region, and insurance charges.

Download the dataset here:

Medical Insurance Dataset Download

PROJECT INSTRUCTION (READ CAREFULLY)

As a data science team, your first responsibility is to understand the business problem clearly.

The company wants to improve how insurance premiums (charges) are calculated for customers. Instead of relying on rough estimates, they want to use data to understand how patient attributes affect insurance cost.

Your task is to build a model that predicts insurance charges based on features such as age, BMI, smoking status, number of children, sex, and region.

You must clearly explain:

- What problem the insurance company is trying to solve
- Why accurate insurance prediction is important
- How better predictions can improve business decisions and fairness

You are expected to load the dataset using Python and carefully inspect it, remember all we did during data inspection in class. At this stage, your goal is simply to understand what data you are working with.

You are expected to clean the dataset using basic techniques only; handle missing values, check and deal with duplicates (if exist) etc. Keep it simple and clean.

The EDA section is very important because it helps you understand patterns in the data. You should perform extensive EDA on your data, keep in mind that it is important for you to explain what each visualization means in simple terms, focusing on what the patterns mean in a healthcare context.

Furthermore, you are expected to demonstrate basic SQL knowledge by loading the dataset into a database. Your SQL analysis should answer simple questions like:

- What is the average insurance charge for smokers vs non-smokers?
- Which age groups have the highest average insurance cost?
- How does BMI category affect average charges?

You are not limited to only the questions above, however, keep SQL simple and focused on understanding data, not complexity.

Before building your model, you must prepare your dataset. You should:

- Separate features (X) from target (y)
- Split data into training and testing sets using train_test_split
- Explain why train-test split is necessary

You are required to build a Linear Regression model only. The goal is to predict insurance charges based on patient features.

You are expected to:

- Train the model using Scikit-learn
- Make predictions on test data
- Understand what the model is doing (not just code it)

Evaluate your model, explain what the accuracy score means, whether the model is performing well or poorly and also what factors might affect the score.

Finally, you must interpret your results in simple language.

FINAL DELIVERABLES

Each group must submit:

Jupyter Notebook (full workflow)

SQL file (basic queries only)

Presentation slides (simple storytelling)

Short written report (findings + recommendations)

FINAL NOTE:

This project is designed to give you real-world experience in healthcare data science using a simple and understandable workflow. The goal is not to build a complex or highly optimized model, but to demonstrate that you understand how to take a dataset, analyze it, build a basic predictive model, and translate results into meaningful business insights.

WISH YOU ALL THE VERY BEST!!